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OWASP
AppSec Europe
London 2nd-6th July 2018

Prepare(): Introducing novel Exploitation Techniques in WordPress

Robin Peraglie



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Prepare(): Introducing no - - PowerPoint PPT Presentation

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Prepare(): Introducing novel Exploitation Techniques in WordPress

Robin Peraglie Prepare(): Introducing

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[\[image: slide-1\]](#)

SLIDE 1

Prepare(): Intr

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Techniques in WordPress

Robin Peraglie

[\[image: slide-2\]](#)

SLIDE 2

- M. Sc. IT-Security @ Ruhr-University Bochum, Germany

Security Researcher @ RIPS Technologies Love breaking stuff with RIPS Code Analysis:

- Moodle RCE
- Prestashop RCE
- LimeSurvey RCE
- CubeCart RCE
- Roundcube RCE

WordPress exploitation (Credits: Slavco Mihajloski and Karim El El Oue uerghemmi)

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Techniques in WordPress

Robin Peraglie

[\[image: slide-3\]](#)

SLIDE 3

- WordPress: open source content management system
- 30% of webhosts run WordPress to create websites blogs and web apps!
- Written in PHP: very flexible but prone to many software vulnerabilities
- Open bugbounty program on Hackerone => hardened core!
- How to exploit?

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Techniques in Word Press

[\[image: slide-4\]](#)

SLIDE 4

- WP core is customized & extended by many great and powerful plugins
- Plugins often bring nasty bugs nullifying security established by bug bounty program
- We will examine design fmaws in WP core that can be exploited through many plugins

Extensibility

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Techniques in Word Press

[\[image: slide-5\]](#)

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Prepare(): Intr

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- # Techniques in WordPress

SLIDE 6

- ```
$wpdb wpdb->que
```

- >query("SELECT ... WHERE name='\$\_GET[0]' "); SQLi not exploitable!
- 4. Custom implementation of Prepared Statements/DBAL

# Security Defense

## Prepare(): Intr

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## Techniques in WordPress

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[\[image: slide-7\]](#)

SLIDE 7

PHP extension PDO offers well-tested "pretty-secure" Prepared Statements PDO::prepare(), PDO::bind(), PDO::execute() Why implement your own? => Legacy code can't be removed: backwards-compatibility between plugins and core! => Switching to PDO would require to rewrite all plugins!

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## Prepared Statements

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## Prepare(): Intr

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## Techniques in WordPress

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[\[image: slide-8\]](#)

SLIDE 8

Very similar to Prepared Statements! Simple use-case: prepare() sanitizes potentially malicious user-input, embeds it in single quotes for placeholders in a SQL

- query. User-input 1'O

'OR'1='='1 would result in a harmless SQL query: SELECT \* FROM table WHERE column1 = '1\OR\1\='1'

## Cust

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## Prepared Statements

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### Prepare(): Intr

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## Techniques in WordPress

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```
$query = $wpdb->prepare("SELECT * FROM table WHERE column1 = %s", $_GET['c1']); $wpdb->query($query);
```

[\[image: slide-9\]](#)

SLIDE 9

## Expl

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**T e c h n i q u e # 1**

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**P r e p a r e ( ) : I n t r**

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**T e c h n i q u e s i n W o r d P r e s s**

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[\[image: slide-10\]](#)

SLIDE 10

WordPress earlier than 4.8.3 was vulnerable to a SQL injection located in this very commonly used code construct known as „do double preparing“.

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**P r e p a r e ( ) : I n t r**

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## using

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## Techniques in WordPress

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```
$query = $wpdb->prepare("SELECT * FROM table WHERE column1 = %s", $_GET['c1']); $query =
$wpdb->prepare($query . " AND column2 = %s", $_GET['c2']); $wpdb->query($query);
```

[\[image: slide-11\]](#)

SLIDE 11

WordPress earlier than 4.8.3 was vulnerable to a SQL injection located in this very commonly used code construct known as „double prepared“. The SQL Injection occurs whenever the user-input contains placeholders!

- input

put contains placeholders! script.php?c1= %s &c2[]=OR 1=1 -- x&c2[]=abc

## D

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## Prepare(): Intr

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## using

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- v

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## Techniques in WordPress

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```
$query = $wpdb->prepare("SELECT * FROM table WHERE column1 = %s", $_GET['c1']); $query = $wpdb->prepare($query . " AND column2 = %s", $_GET['c2']); $wpdb->query($query);
```

[\[image: slide-12\]](#)

SLIDE 12

WordPress earlier than 4.8.3 was vulnerable to a SQL injection located in this very commonly used code construct known as „do double preparing“. The SQL Injection occurs whenever the user-input

- input

contains placeholders! script.php?c1= %s &c2[]=OR 1=1 -- x&c2[]=abc Prepare() #1:  
SELECT \* FROM table WHERE column1 = ' %s '

## D

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## doublePrepare

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## Prepare():Intro

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## Techniques in WordPress

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```
$query = $wpdb->prepare("SELECT * FROM table WHERE column1 = %s", $_GET['c1']); $query = $wpdb->prepare($query . " AND column2 = %s", $_GET['c2']); $wpdb->query($query);
```

[\[image: slide-13\]](#)

SLIDE 13

WordPress earlier than 4.8.3 was vulnerable to a SQL injection located in this very commonly used code construct known as „double prepared“. The SQL Injection occurs whenever the user input contains placeholders!

- input

script.php?c1= %s &c2[]=OR 1=1 -- x&c2[]=abc Prepare() #1:  
SELECT \* FROM table WHERE column1 = ' %s ' AND column2 = %s

## D

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- u

## doublePrepare

---

## Prepare():Intro

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- d

## using

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- v

## example

---

- i

## table

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- n

## Techniques in WordPress

---

```
$query = $wpdb->prepare("SELECT * FROM table WHERE column1 = %s", $_GET['c1']); $query = $wpdb->prepare($query . " AND column2 = %s", $_GET['c2']); $wpdb->query($query);
```

[\[image: slide-14\]](#)

SLIDE 14

WordPress earlier than 4.8.3 was vulnerable to a SQL injection located in this very commonly used code construct known as „double prepared“. The SQL Injection occurs whenever user-input

- input

put contains placeholders! script.php?c1= %s &c2[]=OR 1=1 -- x&c2[]=abc Prepare() #1: SELECT \* FROM table WHERE column1 = ' %s ' AND column2 = %s Prepare() #2: SELECT \* FROM table WHERE column1 = ' 'OR 1=1 -- x' ' AND column2 = 'abc';

## D

---

- u

## doublePrepare

---

## Prepare():Intro

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## usingn

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## Techniques in WordPress

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```
$query = $wpdb->prepare("SELECT * FROM table WHERE column1 = %s", $_GET['c1']); $query = $wpdb->prepare($query . " AND column2 = %s", $_GET['c2']); $wpdb->query($query);
```

[\[image: slide-15\]](#)

SLIDE 15

To mitigate the SQL injection WordPress released a fix for prepare(), which would replace all placeholders in user-input with a unique secret 66-character string before returning from prepare.

## Patch

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## Prepare(): Intr

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## using n

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## Techniques in WordPress

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```
function prepare($query, $args) { if(is_array($args[0])) $args = $args[0]; $query = preg_replace(
'/%s/', ""'%s'", $query); array_walk($args, array($this, 'esc_sql')); $query = vsprintf($query, $args);
return str_replace('%', $this->placeholder_escape(), $query); } function query($query) {
$query=str_replace($this->placeholder_escape(), '%', $query); // send $query to database... }
```

[\[image: slide-16\]](#)

SLIDE 16

With the patch applied all percent signs % in our exploit are effectively replaced with unique secret 66- character string. User-input: script.php?c1= %s &c2[]=abc Prepare() #1: SELECT \* FROM table WHERE column1 = ' {13f...0d23}s ' Prepare() #2: SELECT \* FROM table WHERE column1 = ' {13f...0d23}s ' AND column2 = 'abc'; Query(): SELECT \* FROM table WHERE column1 = ' %s ' AND column2 = 'abc';

## Impact

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## Patch

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## Prepare(): Intr

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## T e c h n i q u e s i n W o r d P r e s s

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```
$query = $wpdb->prepare("SELECT * FROM table WHERE column1 = %s", $_GET['c1']); $query =
$wpdb->prepare($query . " AND column2 = %s", $_GET['c2']); $wpdb->query($query);
```

[\[image: slide-17\]](#)

SLIDE 17

## = > E x p l

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## T e c h n i q u e # 2

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## P r e p a r e ( ) : I n t r

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## e l Expl

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## Techniques in WordPress

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[\[image: slide-18\]](#)

SLIDE 18

The WP\_Query object retrieves wordpress posts from the database which match arguments of constructor `$query_results=new WP_Query('cat=5&post_meta_key=thumbnail');`

## Backgr

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## Prepare():Intr

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## Techniques in WordPress

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DB

`SELECT * FROM wp_posts WHERE ... category=5 and post_meta_key='thumbnail'`  
parsed into executes Results and SQL query stored in WP\_Query!

[image: slide-19]

SLIDE 19

WordPress recommends to cache the results of slow database queries in the database temporarily. Excerpt from the official WordPress Codex manual: `if(false === ($query_results = get_transient('query_results'))){ // cache miss? $query_results=new WP_Query('cat=5&order=random&tag=tech&post_meta_key=thumbnail'); set_transient('query_results', $query_results, 12 * HOUR_IN_SECONDS ); // set cache }` To improve performance the result of the slow database query is cached and omitted in the next run. However, how does the `set_transient()` stores objects in the database?

## Background

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## WordPress Codex

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## Example

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## PreparedStatement

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## Using

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## Example

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## Table

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## Techniques in WordPress

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[image: slide-20]

SLIDE 20



Our WP\_Queue object is stored in \$value function set\_transient( \$transient, \$value, \$expiration = 0)) { \$result = add\_option( \$transient\_option, \$value, '', \$autoload ); } function add\_option( \$option, \$value = '', \$deprecated = '', \$autoload = 'yes' )) { \$serialized\_value = maybe\_serialize( \$value ); \$result = \$wpdb->query(\$wpdb->prepare( "INSERT INTO \$wpdb->options (...) VALUES (%s,%s,%s) ...", ..., \$serialized\_value, ...)); }

## set\_transient()/add\_

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## Prepare():Intr

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## Techniques in WordPress

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[\[image: slide-21\]](#)

SLIDE 21

serialize() translates variable content(strings, arrays, objects,...) to a readable string representation  
unserialize() restores the variable-contents given its serialized string representation.

## Recap:Serializati

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### Prepare():Intr

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## elExpl

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## tati

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## Techniques in WordPress

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\$var serialize(\$var) Integer: \$var = 1; String: \$var = 'hello0WASP'; Array: \$var = array(0=>21,1=>22,23); Object: \$var=new stdClass(); \$var.a="b"; i:1; s:10:"hello0WASP"; a:3: {i:0;i:21;i:1;i:22;i:3;i:23;} O:8:"stdClass":1:{s:1:"a";s:1:"b";}

[\[image: slide-22\]](#)

SLIDE 22

unsanitized user-input reaches un unserialize() => PHP Object injection vulnerability which can cause RCE class LogHandler { public \$file; function \_\_destruct() { file\_put\_contents(\$this->file, "Closing ".\$this->file, FILE\_APPEND); } } unserialize(\$\_GET["p"]); // O:10:"LogHandler":1: {s:4:"file";s:19:"<?=\$\_GET[0]?>.php"} „Magic method“ \_\_destruct() is automatically called if a LogHandler object is removed from memory

## Recap: PHP Object Injecti

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### Prepare():Intr

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## T e c h n i q u e s i n W o r d P r e s s

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[\[image: slide-23\]](#)

SLIDE 23

„WooCommerce“: one of the most popular WordPress plugins with 2.3 million installations  
Affected by exploitation technique 2 by example, leads to authenticated RCE in this case The  
WooCommerce products-shortcode inserts a pretty product-list to a post Attributes can be passed  
to it: [products category=“toasters”]

## T e c h n i q u e 2 : E x a m p l e W o

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## P r e p a r e ( ) : I n t r

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## T e c h n i q u e s i n W o r d P r e s s

---

[image: slide-24]

SLIDE 24

Implementation of products-shortcode as recommended by the WordPress Codex! protected function get\_products() { \$transient\_name = ...; \$products = get\_transient( \$transient\_name ); if ( false === \$products || ! is\_a( \$products, 'WP\_Query' ) ) { \$products = new WP\_Query( \$this->query\_args ); set\_transient( \$transient\_name, \$products, DAY\_IN\_SECONDS \* 30 ); } return \$products; } User-input via shortcode WordPress Codex code construct

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## T e c h n i q u e s i n W o r d P r e s s

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[\[image: slide-25\]](#)

SLIDE 25

[products category="toasters" sku="%"]

## W o

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- C
- mme

## r c e p r

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## Techniques in WordPress

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O:8:"WP\_Query":1:{s:3:"sql";s:100:"SELECT... sku='{a93..dc}""; } property value \$sql SELECT... WHERE... cat=5 sku='{a93..dc}' WP\_Query

- bject

serialize() percent-signs are replaced as introduced in prepare!

[\[image: slide-26\]](#)

SLIDE 26

## set\_transient()

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## Prepare():Intr

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## uc ingn

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## elExpl

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## Techniques in WordPress

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WP\_Query

- bject

serialize() O:8:"WP\_Query":1:{s:3:"sql";s:100:"SELECT... '{a93..dc}'";} DB prepare() query() INSERT INTO ... O:8:"WP\_Query":1:{s:3:"sql";s:100:"SELECT... '{a93..dc}'";} WP\_Query

- bject

INSERT INTO ... O:8:"WP\_Query":1:{s:3:"sql";s:100:"SELECT... '%";} unserialize() O:8:"WP\_Query":1:{... s:100:"SELECT... '%";}

## get\_transient()

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[\[image: slide-27\]](#)

SLIDE 27

## Manipulati

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- n
- f

## serializedrepresentati

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- n

## Prepare():Intr

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## ucingn

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- v

## elExpl

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- n

## TechniquesinWordPress

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O:8:"WP\_Query":1:{s:3:"sql";s:100:"SELECT... sku='% '";...;s:7:"content";s:11:"somecontent";} 35 ≠

[\[image: slide-28\]](#)

SLIDE 28

## Manipulati

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- n
- f

## serializedrepresentati

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- n

## Prepare():Intr

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## ucingn

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- v

## elExpl

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- n

## TechniquesinWordPress

---

O:8:"WP\_Query":1:{s:3:"sql";s:100:"SELECT... sku='% '";...;s:7:"content";s:11:"somecontent";} 35 ≠  
100

[\[image: slide-29\]](#)

SLIDE 29

## Manipulati

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- f



## serializedrepresentati

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- n

## Prepare():Intr

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## ucingn

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## TechniquesinWordPress

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O:8:"WP\_Query":1:{s:3:"sql";s:100:"SELECT... sku='% ' ";...;s:7:"content";s:11:"somecontent";} 35 ≠ 100

[\[image: slide-30\]](#)

SLIDE 30

## Manipulati

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## serializedrepresentati

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## Prepare():Intr

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## ucingn

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## T e c h n i q u e s i n W o r d P r e s s

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O:8:"WP\_Query":1:{s:3:"sql";s:100:"SELECT... sku='% '";...;s:7:"content";s:11:"somecontent";} 35 ≠ 100

[\[image: slide-31\]](#)

SLIDE 31

## M a n i p u l a t i

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## s e r i a l i z e d r e p r e s e n t a t i

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## P r e p a r e ( ) : I n t r

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## Techniques in WordPress

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```
O:8:"WP_Query":1:{s:3:"sql";s:100:"SELECT... sku='%'
";...;s:7:"content";s:11:"some";i:0;O:8:"EvilClass":0:{}i:1;s:0:"";} 35 ≠ 100
```

[\[image: slide-32\]](#)

SLIDE 32

## Manipulation

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- n
- f

## Serialized representation

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- n

## Prepare(): Intr

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- d

## using n

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## elExpl

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## Techniques in WordPress

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```
O:8:"WP_Query":1:{s:3:"sql";s:100:"SELECT... sku='%'
";...;s:7:"content";s:11:"some";i:0;O:8:"EvilClass":0:{}i:1;s:0:"";} 35 ≠ 100 PHP Object Injection!
```

[\[image: slide-33\]](#)

SLIDE 33

## Expl

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## Prepare(): Intr

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## uc ing n

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## Techniques in WordPress

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[\[image: slide-34\]](#)

SLIDE 34

- Unpatched design flaws in WP core
- Lead to exploit techniques against plugins
- In general: Avoid unserialize(), minimize plugin amount
- Code auditors:
- Check for WP\_Query caching
- Check for double prepare
- Check for modified serialized data

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**Techniques in WordPress**

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[\[image: slide-35\]](#)

SLIDE 35

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## Techniques in WordPress

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[\[image: slide-36\]](#)

SLIDE 36

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## viaFileDelete

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## Prepare():Intr

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## **T e c h n i q u e s i n W o r d P r e s s**

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